# ROGER LUO

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## SUMMARY

NUS MSc student (graduating Jan 2027) focused on planning and operations modelling. I build capacity and production-planning models (MILP, discrete-event simulation) in Python and SQL, and currently track material deliveries and supplier testing for a semiconductor fab facade package in Singapore.

## EDUCATION

### National University of Singapore

MSc in Civil Engineering (Transport and Urban Mobility) | Aug 2025 - Jan 2027 (expected)
Relevant: Advanced Construction Planning and Control, Lean Construction, Intermodal Transportation Operations, Transportation Data Analytics; research project (A-)

### Chang'an University

Bachelor of Engineering in Transportation (Excellence Engineer Program) | Sep 2021 - Jun 2025
Relevant: Operations Research, Systems Engineering, Probability and Statistics

## EXPERIENCE

### China State Construction Far East Facade, Singapore

Engineering Intern, Project Engineering | Jun 2026 - Present | Semiconductor fab (Micron) and LTA office facade packages

- Maintain the list, status and delivery plan for 15 facade material types; track milestones on a board that links supplier deliveries to site installation, so late items are flagged early.
- Coordinate sample submission, compliance and performance testing with four suppliers; inspect installation progress and quality on site daily and report deviations, risks and resource gaps at weekly contractor meetings and in monthly reports.
- Wrote Python scripts that parse supplier shipment and test emails into batch and result records, and built a React/Node.js/SQLite goods-receipt app to keep material records consistent.

### Chery International

Product Intern | May 2026 - Jun 2026

- Built the full-cycle Gantt chart for a KD localisation project, defining milestones and the critical path; aligned R&D, production and logistics schedules for sample and production parts.

### Expeditors, Guangzhou

Supply Chain Intern | Dec 2024 - Apr 2025

- Prepared and checked customs declarations; reconciled shipment data with Excel lookups and pivot tables, improving document processing efficiency by about 30%.

## SELECTED PROJECTS

### Production Planning, Downtime Recovery and Lot Sizing

Personal project; synthetic shop and public Trigeiro F1 benchmark | Sep 2026 | manufacturing-planning-agent

- Formulated a two-stage production model with machine capacity, operation precedence, due dates and overtime; solved it with HiGHS. In a 16-unit outage case, the deadline-priority plan kept all units on time with 1 overtime hour, versus 3 under an earliest-due-date rule.
- Replayed plans in SimPy with processing-time variation and extra downtime, then replanned on the updated capacity. Solved the Trigeiro F1 lot-sizing benchmark (6 products, 15 periods) at cost 40,854.

### Registration Data Pipeline and SQL Query Service

Personal project; official public data | Sep 2026 | malaysia-auto-ask

- Validated 1,436,804 records in Python, built Hive tables and 1,536 aggregates, and loaded them into MySQL with versioned imports that roll back on failure; cross-checked 130 query results.

### AYE Expressway Bottleneck Analysis

NUS CE5203 group project | Spring 2026 | ce5203-aye-weaving

- Turned YOLOv11 vehicle counts into a calibrated SUMO model of a 373 m weaving section; across 12 scenarios, ramp metering cut peak network time loss by 22.7%, while speed limits made it worse.

### Civil Buddy: AI Workbench for Engineering SMEs

Team of 4 | 3rd place, SME category, NUS-ISS "Show Me Your Agents" Hackathon 2026 (114 teams) | civil-buddy-sme

## SKILLS

**Planning and modelling:** capacity and production planning, scheduling, lot sizing, LP/MIP (HiGHS), discrete-event simulation (SimPy), SUMO.

**Data and tools:** Python (pandas, NumPy, SciPy), SQL (MySQL, Hive), Excel (pivot tables, lookups), Git, React/Node.js; AI-assisted coding and analysis.

**Languages:** English (professional), Mandarin (native).